

To what extent can weather conditions be used to predict flight delays?

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Introduction - Claire

This project tackles a simple, practical question: To what extent can weather conditions be used to predict flight delays? We focus on Florida in January 2025, a time and place where storms and heavy rain are common enough to matter. Our dataset combines flight records with daily weather at both the origin and destination airports. The weather comes from the Open-Meteo archive and arrives as single-value NumPy arrays (for example, [0.]). We convert those arrays into plain numbers and keep the columns easy to follow by adding `_ORIGIN` and `_DEST` to show where each weather reading came from.

The plan is to get the data into credible shape, look for patterns, and then test if those patterns have any predictive power. We start by exploring the datasets, removing missing values, and narrowing the scope to Florida in January. If the full table is still too big, we work with a sensible subset so analysis and modelling stay fast. From there, we create clear visuals—histograms, and correlation heatmaps—to understand how delay measures relate to weather features such as wind, rain, and daylight at departure and arrival.

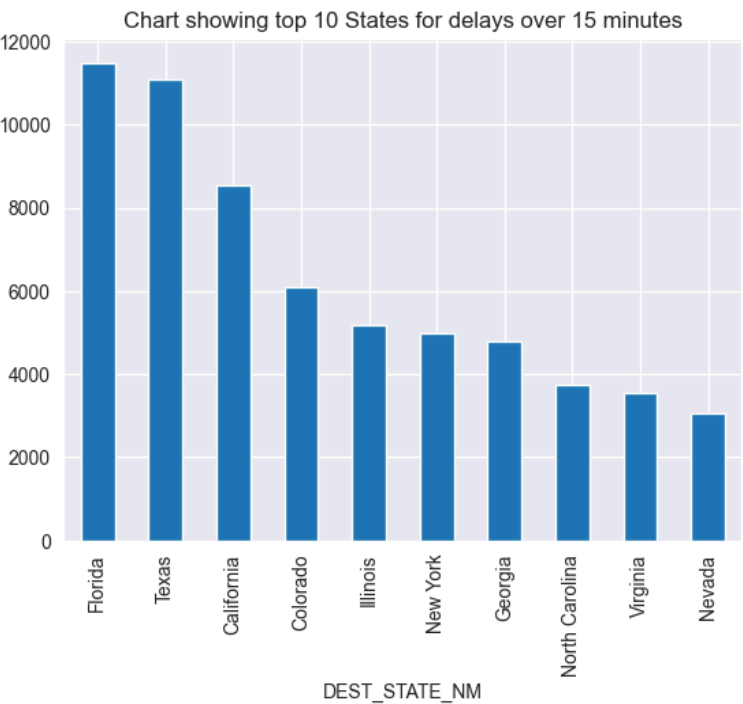
With the groundwork done, we build a straightforward baseline model. We split the data into training and test sets and train a decision tree classifier to predict a simple class label: delayed or not delayed (for example, using `WEATHER_BINARY`). We then make predictions on the test set, report the accuracy, and show a confusion matrix so it's obvious where the model succeeds and where it trips up. The aim isn't just to produce a number—it's to see whether weather alone carries enough signal to be useful, and to be honest about its limits.

The report is organised as follows: we begin with the Background, setting the context and explaining why Florida in January 2025 is a useful test bed. We then move to Specification & Design, which lays out the requirements and the high-level pipeline (data preparation, joining weather with `_ORIGIN/_DEST`, visual exploration, and a light predictive check). Implementation & Execution describes how the work was carried out in practice—tools, roles, key decisions, and challenges. Data Collection appears near the end and documents the sources and how we obtained and merged them (BTS flights, Open-Meteo weather, and airports/Kaggle). We finish with the Conclusion, which summarises what we learned about how far weather alone can predict delays and the most sensible next steps.

Background - Maya

Flight delays are a frequent issue in the aviation industry, affecting airlines, airports, and passengers alike. Delays result in significant costs for all parties: airlines face compensation payouts and a loss of passenger trust, airports struggle with congestion and frustrated customers and passengers experience missed connections and disrupted holidays.

While many factors contribute to delays, extreme weather is one of the hardest to predict. Storms, hurricanes and heavy rainfall can bring air travel to a halt. With climate change making extreme weather more frequent, it's becoming more critical to understand its impact and whether delays caused by weather can be predicted.



Predicting weather-related delays could bring significant benefits. Airlines and airports can't control the weather, but having reliable forecasts of potential disruptions would allow them to plan flight routes more effectively, manage resources and minimise operational challenges. For passengers, this would mean clearer explanations for delays (leading to less frustration directed at airlines and airports) and a better understanding of when they may be eligible for compensation.

Due to the wealth of data available, we chose to focus on Florida as it regularly experiences extreme weather, including hurricanes, thunderstorms and heavy rainfall, making it an ideal test case. By combining flight records with weather datasets, we aim to investigate whether these factors can be used to predict delays with reasonable accuracy.

Our project aims to uncover patterns and build models that predict the likelihood of a flight being delayed due to weather conditions. While this study focuses on Florida as a starting point, the approach could be extended to regions with less severe weather to test how accurately we can predict these types of delays.

Specification & Design: Riya

This section outlines the specifications and design of our data science project. We define the key requirements, both technical and non-technical, that we followed during the development process, ensuring the project meets its objectives while remaining usable. We also present the overall system design and architecture, describing how data will flow from collection through preprocessing, analysis, modelling, and visualisation.

1. Requirements:

1.1 Technical Requirements:

- Data Acquisition: Import data from multiple CSVs (eg: IATA airport codes and longitude and latitude values (airports.csv)) and API (Historical Weather API (open-meteo) – Daily weather data for origin and destination airport)
- Data Preprocessing: Handling missing values, outliers, categorical encoding (for machine learning, using one-hot encoding) and normalisation.
- Exploratory Data Analysis: correlation heatmaps, feature importance and statistical analysis
- Modelling: Training 2 machine learning models (Logistic Regression and Random Forest) to answer what factors affect weather delays.
- Model Evaluation: Evaluate models using metrics such as accuracy, precision, recall and F1-Score from the classification tables that were generated.
- Visualisation: Present results through charts using Matplotlib and Seaborn
- Collaboration: Code and analysis maintained via GitHub and Google Drive for version control and documentations.

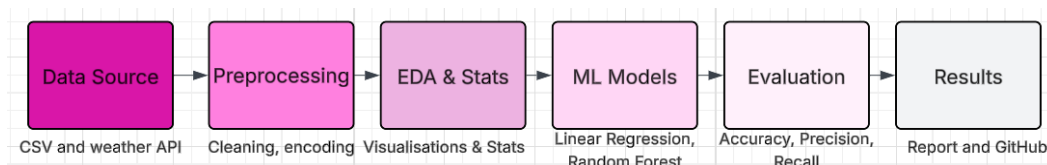
1.2 Non-Technical Requirements:

- Usability: Visualisations and results must be easily understood by non-technical users. In this case, the project is focused on a wide range of stakeholders such as airline companies and airline customers, so the findings should be easily understood by all of them.
- Time Constraints: Project completed in 5 weeks by 6 team members, delivering on time.
- Scalability: The design should allow extending to new datasets
- Impact: Deliver insights that are meaningful to the target stakeholders (airline customers, airline companies, etc.)

2. Design & Architecture

2.1 System Architecture Design:

The system architecture shows the workflow of the project, showing how the data moves through different stages of processing. Raw data is collected and preprocessed to handle missing values, outliers and to do feature encoding. The cleaned dataset is explored through statistics and visualisation, before being trained and evaluated through 2 ML models. The models are then evaluated using various metrics and the final results are explained through the GitHub README and report.



2.2 Technology Stack:

- Programming Language: Python
- Libraries: Pandas, NumPy, skikit-learn, Matplotlib, Seaborn
- Collaboration Tools: GitHub, Google Drive, Jupyter Notebook
- Data: CSV files and API calls.

Implementation & Execution - Danielle

We adopted a collaborative approach. During the brainstorming stage, we began by exploring our interests and identifying any we had in common, before holding a vote on which to do our project on. Each member of the team then took responsibility for part of the project to work on individually while maintaining regular check-ins through communication on Slack and team meetings.

Responsibilities included:

Danielle - exploration of flights and flights delay data set. Combining relevant data sets, exploration and data cleaning, assisting with API calls.

Heather - writing code for API call and running to combine into one data set

Laura - finding data sources, exploratory analysis on climate data

Riya - finding data sources, machine learning

Maya - machine learning

Claire - exploratory analysis on Kaggle data set and Florida delays weather data

We primarily worked in Python using Jupyter Notebook. Key libraries included:

- Pandas for data cleaning, merging, and manipulation
- NumPy for numerical operations
- Matplotlib and seaborn for visualisation
- Scikit-learn for machine learning and model evaluation

Our achievements included successfully merging multiple data sets to include the variables we needed to call the API and our target variable. We created clear visualisations of key trends, and implemented predictive modelling. Furthermore, we managed to call the weather API for all of the Florida flights once we had filtered to a manageable

size to ensure accurate data. API results were stored to avoid hitting limits and to allow reproducibility without re-calling the API.

One of the main challenges was the size and complexity of the dataset. The raw data contained a very large number of records and attributes, many of which were not directly relevant to our research questions. To make the analysis manageable, we had to carefully filter, sample, and prioritise key variables. We did this by selecting Florida as our location as through our exploratory analysis, we identified that there was interesting weather variation here.

At several points, we revised our methodology—for instance, adjusting from using the airline delay cause data set to the flights delay data set, which contained information on a flight-by-flight basis. This was done as our initial data set had the data on all flights within a given month combined from each airport, with no way of identifying which flight was delayed for each reason. We adapted by downloading a new data set from the source which included data on a flight-by-flight basis.

While we did not follow a formal Agile framework, we did implement iterative improvements when we revisited code after feedback from team members. We also restructured code to improve efficiency, particularly when working with the API.

Data Collection - Heather

In order to answer our project question, we needed to find data sources related to flight delays and the weather, ideally with both pieces of information combined in the same dataset. Fortunately, we were able to find a number of different data sources that at first look appeared to be relevant to our project, these included data from the National Centers for Environmental Information, the Historical Weather API (open-meteo), the Bureau of Transport Statistics (USA) and Kaggle. After careful review of all the datasets, we realised that there was a wealth of information related to the USA, so we decided to make this our focus.

The Historical Weather API from open-meteo was targeted as a resource to retrieve daily weather information. The API requires passing date, longitude and latitude values as parameters when performing a GET request. We noticed that we had airport IATA codes along with airport longitude and latitude values in one of the Kaggle datasets and started to see a connection between the two sources of data. Additionally, in Kaggle, was a dataset that included flight date, origin airport and destination airport IATA codes, flight departure and arrival delay in minutes and as binary classes, and weather delay in minutes. At this point, we realised there was a link between all 3 datasets and combining the three of them would allow us to combine flight delay data with historical weather information. The flight delays data from Kaggle is from 2015 and after some searching online, we discovered that the origin of the data was the Bureau of Transport Statistics and that we could download more recent data of the same type, resulting in the following 3 data sources being used in the project:

Kaggle – IATA airport codes and longitude and latitude values (airports.csv)

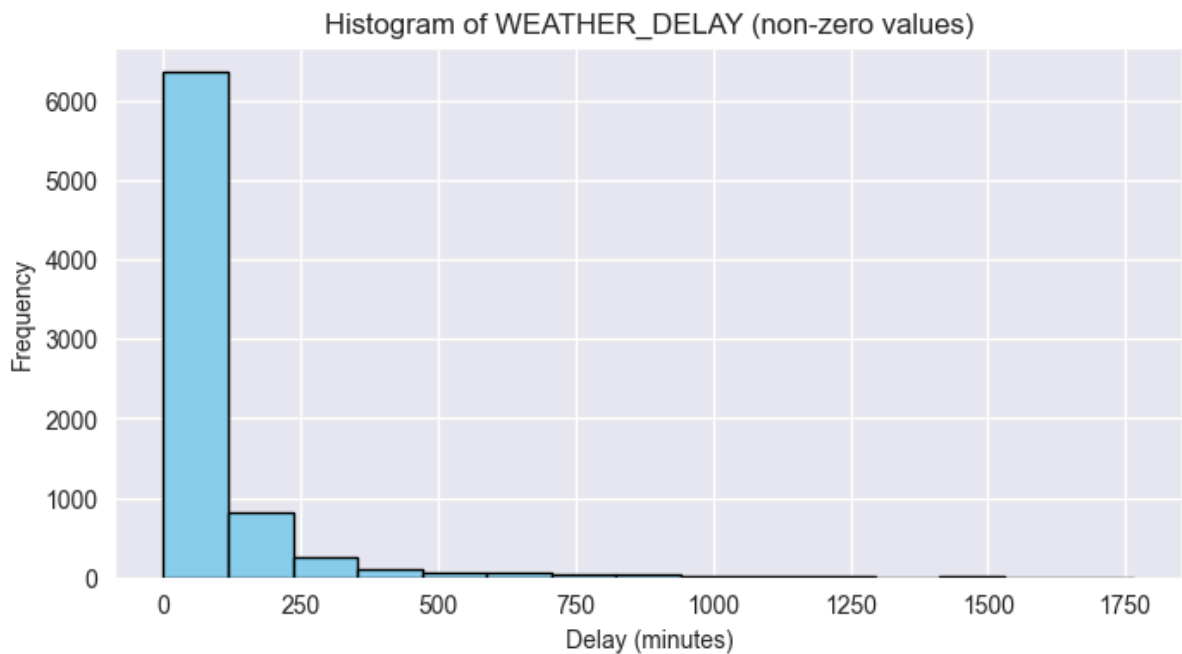
Bureau of Transport Statistics – flight delays data for January 2025 for internal flights in the USA (delays_by_flight.csv)

Historical Weather API (open-meteo) – Daily weather data for origin and destination airports (<https://open-meteo.com/en/docs/historical-weather-api>)

The process of combining the data sources can be seen in the main Jupyter Notebook. Initially, the airports and flight delays datasets were read and merged for both origin and destination. The dataset of flights has information for over 500000 flights and with a daily API limit of 10000 calls, we knew this was not feasible in our timeframe. Therefore, we filtered the flights dataset removing any rows with null values, kept only flights with an origin in Florida and filtered to flights we knew were delayed. Therefore, leaving us with just under 10000 rows of data,

more realistic for the time and API constraints of the project. After filtering to only delayed flights, we now had a dataset of flights we knew were delayed and a column that showed the number of minutes they were delayed by the weather (0 meaning no minutes delayed due to the weather). Therefore, a binary column called 'WEATHER_BINARY' was created (0 = not delayed due to weather and 1 = delayed due to weather) to be the focus of our machine learning tasks.

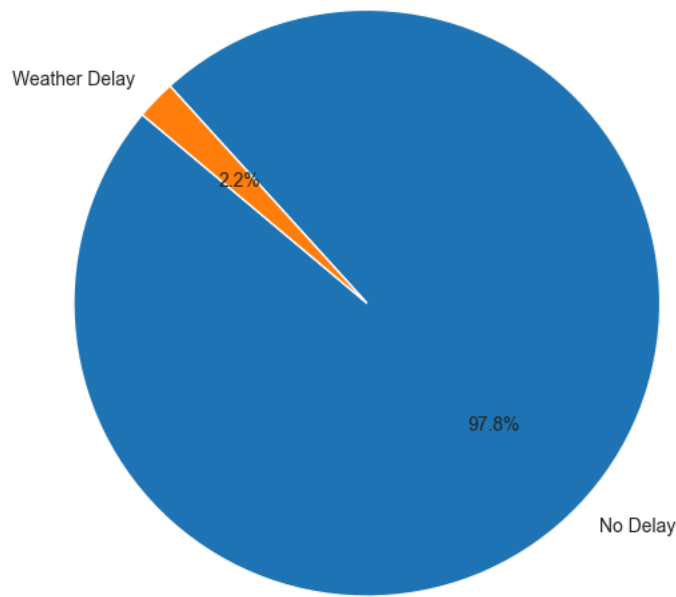
Source: Bureau of Transport Statistics (USA)



Conclusion - Laura

Our project set out to address a straightforward but important question: *to what extent can weather conditions be used to predict flight delays?* To explore this, we combined Bureau of Transport Statistics flight data with historical weather records from Open-Meteo, focusing specifically on Florida in January 2025.

Proportion of Flights With/Without Weather Delay

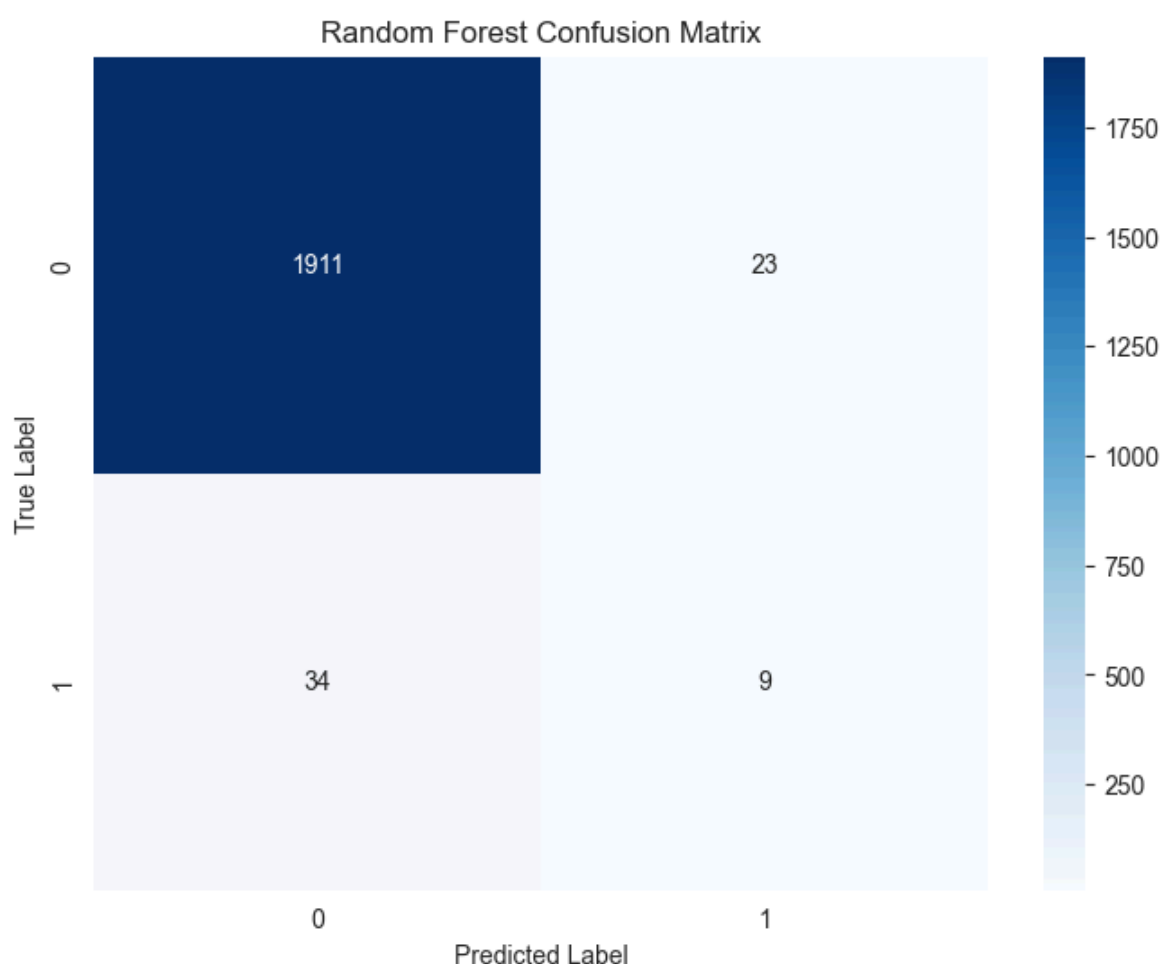


Most of the flights in our dataset—about 97.8%—weren't delayed by weather. Because of this, our models became very good at predicting when a flight wouldn't be delayed, which led to a high number of correct predictions for non-delayed flights (True Negatives).

However, this means we didn't have enough examples of weather-delayed flights for the models to learn from. As a result, while the models excelled at spotting flights that weren't delayed, they struggled to correctly identify the much rarer weather delays. Therefore, even though the accuracy looks high, it doesn't really reflect how well the models can predict the delays that matter most. To get better at forecasting weather-related delays, we would need to address this imbalance perhaps by

resampling the data or trying different modelling techniques.

This outcome reinforces an important point: weather is only one factor among many that influence flight operations. Airline scheduling practices, crew and maintenance availability, air traffic control constraints, and broader network effects all contribute to whether a flight departs or arrives on time. Examining weather in isolation provides a signal, but not a comprehensive explanation.



Despite this, our project still demonstrated several worthy contributions:

1. Our findings show that integrating flight and weather datasets is both feasible and informative.
2. We established that weather features alone can function as a meaningful baseline in predictive modelling.
3. Finally, we produced a framework that can be expanded (by incorporating operational and airline variables) into a more robust forecasting tool.

In response to the research question, we conclude that weather conditions can help predict flight delays, but only to a limited extent. Reliable delay forecasting will require models that integrate larger sets of weather data with operational data.

Our study represents an initial step, pointing toward a more robust predictive system that could ultimately support decision-making by airlines, airports, and passengers alike.